

Table D.3 Superheated Vapor (Steam)*

Table D.3A Isobaric Data for $P = 0.006$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
309.31	0.006	0.042135	23.733	2424.2	2566.6	8.3290
320	0.006	0.040708	24.565	2439.7	2587.1	8.3940
340	0.006	0.038291	26.116	2468.4	2625.1	8.5092
360	0.006	0.036151	27.662	2497.0	2663.0	8.6176
380	0.006	0.034239	29.206	2525.7	2701.0	8.7202
400	0.006	0.032522	30.748	2554.6	2739.0	8.8179
420	0.006	0.030969	32.290	2583.5	2777.3	8.9112
440	0.006	0.029559	33.831	2612.7	2815.7	9.0005
460	0.006	0.028272	35.371	2642.1	2854.3	9.0863
480	0.006	0.027092	36.911	2671.6	2893.1	9.1689
500	0.006	0.026008	38.450	2701.4	2932.1	9.2485
520	0.006	0.025006	39.990	2731.4	2971.4	9.3255
540	0.006	0.024079	41.529	2761.7	3010.9	9.4000
560	0.006	0.023219	43.068	2792.2	3050.6	9.4722
580	0.006	0.022418	44.607	2822.9	3090.6	9.5424
600	0.006	0.02167	46.146	2853.9	3130.8	9.6105
620	0.006	0.020971	47.685	2885.1	3171.2	9.6769
640	0.006	0.020315	49.224	2916.6	3211.9	9.7415
660	0.006	0.019699	50.763	2948.3	3252.9	9.8046
680	0.006	0.01912	52.301	2980.4	3294.2	9.8661
700	0.006	0.018573	53.840	3012.6	3335.7	9.9263
720	0.006	0.018057	55.379	3045.2	3377.4	9.9851
740	0.006	0.017569	56.917	3078.0	3419.5	10.043
760	0.006	0.017107	58.456	3111.0	3461.8	10.099
780	0.006	0.016668	59.995	3144.4	3504.3	10.154
800	0.006	0.016251	61.533	3178.0	3547.2	10.209

*Property values generated from NIST Database 23: REFPROP Version 7.0 (August 2002).

Table D.3B Isobaric Data for $P = 0.035$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
345.83	0.035	0.22099	4.5251	2472.3	2630.7	7.7146
360	0.035	0.21197	4.7176	2493.5	2658.6	7.7939
380	0.035	0.20052	4.9871	2523.1	2697.7	7.8994
400	0.035	0.1903	5.2549	2552.5	2736.5	7.9989
420	0.035	0.1811	5.5218	2581.9	2775.2	8.0934
440	0.035	0.17278	5.7879	2611.4	2813.9	8.1835
460	0.035	0.16519	6.0535	2640.9	2852.8	8.2699
480	0.035	0.15826	6.3187	2670.7	2891.8	8.353
500	0.035	0.15189	6.5837	2700.6	2931.0	8.433
520	0.035	0.14602	6.8484	2730.7	2970.4	8.5102
540	0.035	0.14059	7.113	2761.1	3010.0	8.5849
560	0.035	0.13555	7.3775	2791.6	3049.8	8.6573
580	0.035	0.13086	7.6419	2822.4	3089.9	8.7276
600	0.035	0.12648	7.9062	2853.4	3130.1	8.7958
620	0.035	0.12239	8.1704	2884.7	3170.7	8.8623
640	0.035	0.11856	8.4346	2916.2	3211.4	8.927
660	0.035	0.11496	8.6987	2948.0	3252.5	8.9901
680	0.035	0.11157	8.9627	2980.0	3293.7	9.0517
700	0.035	0.10838	9.2268	3012.3	3335.3	9.1119
720	0.035	0.10537	9.4908	3044.9	3377.1	9.1708
740	0.035	0.10251	9.7547	3077.7	3419.1	9.2284
760	0.035	0.099813	10.019	3110.8	3461.4	9.2848
780	0.035	0.097251	10.283	3144.2	3504.0	9.3402
800	0.035	0.094818	10.547	3177.8	3546.9	9.3944
820	0.035	0.092503	10.81	3211.7	3590.1	9.4477
840	0.035	0.090299	11.074	3245.9	3633.5	9.5

Table D.3C Isobaric Data for $P = 0.070$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
363.08	0.07	0.42287	2.3648	2493.9	2659.4	7.479
380	0.07	0.40301	2.4813	2519.9	2693.5	7.5709
400	0.07	0.38205	2.6175	2550.0	2733.2	7.6727
420	0.07	0.3633	2.7525	2579.9	2772.6	7.7687
440	0.07	0.3464	2.8868	2609.7	2811.8	7.8599
460	0.07	0.33106	3.0206	2639.6	2851.0	7.9471
480	0.07	0.31706	3.154	2669.5	2890.3	8.0307
500	0.07	0.30422	3.2871	2699.6	2929.7	8.1111
520	0.07	0.2924	3.42	2729.9	2969.3	8.1886
540	0.07	0.28147	3.5528	2760.3	3009.0	8.2636
560	0.07	0.27134	3.6854	2790.9	3048.9	8.3362
580	0.07	0.26193	3.8179	2821.8	3089.0	8.4066
600	0.07	0.25314	3.9503	2852.9	3129.4	8.475
620	0.07	0.24494	4.0827	2884.2	3170.0	8.5416
640	0.07	0.23725	4.215	2915.8	3210.8	8.6064
660	0.07	0.23003	4.3472	2947.6	3251.9	8.6696
680	0.07	0.22324	4.4794	2979.6	3293.2	8.7312
700	0.07	0.21685	4.6116	3012.0	3334.8	8.7915
720	0.07	0.2108	4.7437	3044.5	3376.6	8.8504
740	0.07	0.20509	4.8758	3077.4	3418.7	8.9081
760	0.07	0.19968	5.0079	3110.5	3461.1	8.9645
780	0.07	0.19455	5.14	3143.9	3503.7	9.0199
800	0.07	0.18968	5.2721	3177.5	3546.6	9.0742
820	0.07	0.18504	5.4041	3211.5	3589.7	9.1275
840	0.07	0.18063	5.5361	3245.7	3633.2	9.1798
860	0.07	0.17643	5.6681	3280.1	3676.9	9.2313

Table D.3D Isobaric Data for $P = 0.100$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
372.76	0.1	0.59034	1.6939	2505.6	2674.9	7.3588
380	0.1	0.57824	1.7294	2517.0	2689.9	7.3986
400	0.1	0.54761	1.8261	2547.8	2730.4	7.5025
420	0.1	0.52038	1.9217	2578.2	2770.3	7.5999
440	0.1	0.49592	2.0165	2608.3	2810.0	7.6921
460	0.1	0.47378	2.1107	2638.4	2849.5	7.7799
480	0.1	0.45361	2.2045	2668.5	2889.0	7.864
500	0.1	0.43514	2.2981	2698.7	2928.6	7.9447
520	0.1	0.41815	2.3915	2729.1	2968.2	8.0226
540	0.1	0.40247	2.4847	2759.6	3008.1	8.0978
560	0.1	0.38794	2.5777	2790.3	3048.1	8.1705
580	0.1	0.37444	2.6707	2821.3	3088.3	8.2411
600	0.1	0.36185	2.7635	2852.4	3128.8	8.3096
620	0.1	0.3501	2.8563	2883.8	3169.4	8.3763
640	0.1	0.33909	2.9491	2915.4	3210.3	8.4411
660	0.1	0.32876	3.0418	2947.2	3251.4	8.5044
680	0.1	0.31904	3.1344	2979.3	3292.8	8.5661
700	0.1	0.30988	3.227	3011.7	3334.4	8.6264
720	0.1	0.30124	3.3196	3044.3	3376.2	8.6854
740	0.1	0.29307	3.4122	3077.1	3418.3	8.7431
760	0.1	0.28533	3.5047	3110.3	3460.7	8.7996
780	0.1	0.27799	3.5972	3143.6	3503.4	8.855
800	0.1	0.27102	3.6897	3177.3	3546.3	8.9093
820	0.1	0.2644	3.7822	3211.3	3589.5	8.9626
840	0.1	0.25809	3.8747	3245.5	3632.9	9.015
860	0.1	0.25207	3.9671	3280.0	3676.7	9.0665
880	0.1	0.24633	4.0595	3314.7	3720.7	9.1171

Table D.3E Isobaric Data for $P = 0.150$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg · K)
384.5	0.15	0.8626	1.1593	2519.2	2693.1	7.223
400	0.15	0.82612	1.2105	2544.0	2725.6	7.3058
420	0.15	0.78408	1.2754	2575.2	2766.5	7.4057
440	0.15	0.74658	1.3394	2605.9	2806.8	7.4995
460	0.15	0.71278	1.403	2636.4	2846.9	7.5884
480	0.15	0.6821	1.4661	2666.9	2886.8	7.6733
500	0.15	0.65407	1.5289	2697.3	2926.6	7.7547
520	0.15	0.62835	1.5915	2727.8	2966.6	7.833
540	0.15	0.60463	1.6539	2758.5	3006.6	7.9086
560	0.15	0.58268	1.7162	2789.4	3046.8	7.9816
580	0.15	0.5623	1.7784	2820.4	3087.1	8.0524
600	0.15	0.54333	1.8405	2851.6	3127.7	8.1212
620	0.15	0.52562	1.9025	2883.1	3168.4	8.188
640	0.15	0.50903	1.9645	2914.7	3209.4	8.253
660	0.15	0.49348	2.0264	2946.6	3250.6	8.3164
680	0.15	0.47886	2.0883	2978.8	3292.0	8.3782
700	0.15	0.46508	2.1502	3011.1	3333.7	8.4386
720	0.15	0.45209	2.212	3043.8	3375.6	8.4976
740	0.15	0.4398	2.2738	3076.7	3417.7	8.5554
760	0.15	0.42817	2.3355	3109.8	3460.2	8.6119
780	0.15	0.41714	2.3973	3143.3	3502.9	8.6674
800	0.15	0.40667	2.459	3177.0	3545.8	8.7217
820	0.15	0.39671	2.5207	3210.9	3589.0	8.7751
840	0.15	0.38724	2.5824	3245.1	3632.5	8.8275
860	0.15	0.3782	2.6441	3279.7	3676.3	8.879
880	0.15	0.36958	2.7058	3314.4	3720.3	8.9296
900	0.15	0.36135	2.7674	3349.5	3764.6	8.9794

Table D.3F Isobaric Data for $P = 0.300$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg · K)
406.67	0.3	1.6508	0.60576	2543.2	2724.9	6.9916
420	0.3	1.5906	0.62868	2565.8	2754.4	7.063
440	0.3	1.5101	0.6622	2598.4	2797.1	7.1623
460	0.3	1.4387	0.69507	2630.3	2838.8	7.255
480	0.3	1.3746	0.72749	2661.7	2879.9	7.3426
500	0.3	1.3165	0.75958	2692.9	2920.8	7.426
520	0.3	1.2635	0.79144	2724.0	2961.5	7.5057
540	0.3	1.2149	0.82312	2755.2	3002.1	7.5824
560	0.3	1.17	0.85467	2786.4	3042.8	7.6564
580	0.3	1.1285	0.8861	2817.7	3083.6	7.7279
600	0.3	1.09	0.91744	2849.2	3124.4	7.7973
620	0.3	1.0541	0.94871	2880.9	3165.5	7.8646
640	0.3	1.0205	0.97992	2912.7	3206.7	7.93
660	0.3	0.98904	1.0111	2944.8	3248.1	7.9937
680	0.3	0.95951	1.0422	2977.1	3289.7	8.0558
700	0.3	0.93173	1.0733	3009.6	3331.6	8.1165
720	0.3	0.90553	1.1043	3042.4	3373.6	8.1757
740	0.3	0.88079	1.1353	3075.3	3416.0	8.2337
760	0.3	0.85738	1.1663	3108.6	3458.5	8.2904
780	0.3	0.8352	1.1973	3142.1	3501.3	8.346
800	0.3	0.81415	1.2283	3175.9	3544.3	8.4005
820	0.3	0.79414	1.2592	3209.9	3587.6	8.4539
840	0.3	0.7751	1.2901	3244.2	3631.2	8.5064
860	0.3	0.75696	1.3211	3278.7	3675.1	8.558
880	0.3	0.73966	1.352	3313.6	3719.2	8.6087
900	0.3	0.72314	1.3829	3348.7	3763.5	8.6586
920	0.3	0.70734	1.4138	3384.1	3808.2	8.7076

Table D.3G Isobaric Data for $P = 0.50$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
424.98	0.5	2.668	0.37481	2560.7	2748.1	6.8207
440	0.5	2.5579	0.39095	2587.6	2783.1	6.9015
460	0.5	2.429	0.41169	2621.6	2827.4	7.0001
480	0.5	2.3153	0.43191	2654.5	2870.5	7.0917
500	0.5	2.2135	0.45176	2686.8	2912.7	7.1779
520	0.5	2.1215	0.47136	2718.8	2954.5	7.2599
540	0.5	2.0376	0.491	2750.6	2996.0	7.3382
560	0.5	1.9607	0.510	2782.4	3037.4	7.4134
580	0.5	1.8898	0.529	2814.1	3078.7	7.486
600	0.5	1.8242	0.548	2846.0	3120.1	7.5561
620	0.5	1.7631	0.567	2878.0	3161.5	7.6241
640	0.5	1.7063	0.586	2910.1	3203.1	7.6901
660	0.5	1.6531	0.605	2942.4	3244.8	7.7542
680	0.5	1.6032	0.624	2974.8	3286.7	7.8168
700	0.5	1.5564	0.643	3007.5	3328.8	7.8777
720	0.5	1.5123	0.661	3040.4	3371.1	7.9373
740	0.5	1.4706	0.680	3073.6	3413.5	7.9955
760	0.5	1.4313	0.699	3106.9	3456.3	8.0524
780	0.5	1.394	0.717	3140.5	3499.2	8.1082
800	0.5	1.3587	0.736	3174.4	3542.4	8.1629
820	0.5	1.3252	0.755	3208.5	3585.8	8.2165
840	0.5	1.2932	0.77325	3242.9	3629.5	8.2691
860	0.5	1.2629	0.79186	3277.5	3673.4	8.3208
880	0.5	1.2339	0.81045	3312.4	3717.6	8.3716
900	0.5	1.2062	0.82904	3347.6	3762.1	8.4216
920	0.5	1.1798	0.84762	3383.0	3806.8	8.4708
940	0.5	1.1545	0.86619	3418.7	3851.8	8.5191

Table D.3H Isobaric Data for $P = 0.70$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
438.1	0.7	3.666	0.27277	2571.8	2762.8	6.7071
440	0.7	3.6453	0.27433	2575.5	2767.6	6.718
460	0.7	3.4477	0.29005	2612.3	2815.3	6.8242
480	0.7	3.2775	0.30511	2647.0	2860.5	6.9204
500	0.7	3.1274	0.31976	2680.5	2904.3	7.0099
520	0.7	2.9929	0.33413	2713.4	2947.3	7.0941
540	0.7	2.8712	0.34828	2745.9	2989.7	7.1742
560	0.7	2.7603	0.36228	2778.3	3031.9	7.2508
580	0.7	2.6585	0.37616	2810.5	3073.8	7.3244
600	0.7	2.5645	0.38994	2842.7	3115.7	7.3953
620	0.7	2.4775	0.40364	2875.0	3157.6	7.464
640	0.7	2.3965	0.41728	2907.4	3199.5	7.5306
660	0.7	2.3209	0.43086	2939.9	3241.5	7.5952
680	0.7	2.2502	0.4444	2972.6	3283.7	7.6582
700	0.7	2.1839	0.4579	3005.4	3326.0	7.7195
720	0.7	2.1215	0.47137	3038.5	3368.5	7.7793
740	0.7	2.0626	0.48481	3071.8	3411.1	7.8378
760	0.7	2.0071	0.49823	3105.3	3454.0	7.895
780	0.7	1.9545	0.51163	3139.0	3497.1	7.9509
800	0.7	1.9047	0.52501	3172.9	3540.4	8.0058
820	0.7	1.8575	0.53837	3207.1	3584.0	8.0595
840	0.7	1.8125	0.55172	3241.6	3627.8	8.1123
860	0.7	1.7697	0.56505	3276.3	3671.8	8.1641
880	0.7	1.729	0.57838	3311.3	3716.1	8.215
900	0.7	1.6901	0.59169	3346.5	3760.7	8.2651
920	0.7	1.6529	0.60499	3382.0	3805.5	8.3143
940	0.7	1.6174	0.61829	3417.7	3850.5	8.3628

Table D.3I Isobaric Data for $P = 1.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
453.03	1	5.145	0.19436	2582.7	2777.1	6.585
460	1	5.0376	0.19851	2597.0	2795.5	6.6253
480	1	4.7658	0.20983	2634.9	2844.7	6.7301
500	1	4.5323	0.22064	2670.6	2891.2	6.825
520	1	4.3268	0.23112	2705.0	2936.1	6.9131
540	1	4.1431	0.24137	2738.7	2980.1	6.996
560	1	3.9771	0.25144	2772.0	3023.4	7.0748
580	1	3.8258	0.26138	2804.9	3066.3	7.1501
600	1	3.6871	0.27122	2837.7	3109.0	7.2224
620	1	3.5591	0.28097	2870.5	3151.5	7.2921
640	1	3.4404	0.29066	2903.3	3194.0	7.3596
660	1	3.33	0.3003	2936.2	3236.5	7.425
680	1	3.227	0.30988	2969.2	3279.1	7.4885
700	1	3.1305	0.31943	3002.3	3321.7	7.5504
720	1	3.04	0.32895	3035.6	3364.5	7.6107
740	1	2.9547	0.33844	3069.1	3407.5	7.6695
760	1	2.8744	0.3479	3102.7	3450.6	7.727
780	1	2.7984	0.35734	3136.6	3494.0	7.7833
800	1	2.7265	0.36677	3170.7	3537.5	7.8384
820	1	2.6583	0.37618	3205.1	3581.2	7.8924
840	1	2.5936	0.38557	3239.6	3625.2	7.9454
860	1	2.532	0.39495	3274.4	3669.4	7.9974
880	1	2.4733	0.40432	3309.5	3713.8	8.0484
900	1	2.4174	0.41367	3344.8	3758.5	8.0986
920	1	2.3639	0.42302	3380.4	3803.4	8.148
940	1	2.3129	0.43236	3416.3	3848.6	8.1966
960	1	2.264	0.4417	3452.4	3894.1	8.2444

Table D.3J Isobaric Data for $P = 1.5$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
471.44	1.5	7.5924	0.13171	2593.4	2791.0	6.443
480	1.5	7.3885	0.13534	2612.3	2815.3	6.4942
500	1.5	6.9775	0.14332	2652.6	2867.6	6.6009
520	1.5	6.629	0.15085	2690.1	2916.4	6.6967
540	1.5	6.3249	0.1581	2726.1	2963.2	6.7851
560	1.5	6.0549	0.16515	2761.0	3008.8	6.8678
580	1.5	5.8121	0.17206	2795.3	3053.4	6.9462
600	1.5	5.5915	0.17884	2829.2	3097.5	7.0209
620	1.5	5.3897	0.18554	2862.9	3141.2	7.0925
640	1.5	5.2039	0.19216	2896.4	3184.7	7.1616
660	1.5	5.0319	0.19873	2929.9	3228.0	7.2282
680	1.5	4.8721	0.20525	2963.4	3271.3	7.2929
700	1.5	4.7231	0.21173	2997.0	3314.6	7.3556
720	1.5	4.5836	0.21817	3030.7	3358.0	7.4167
740	1.5	4.4527	0.22458	3064.5	3401.4	7.4762
760	1.5	4.3295	0.23097	3098.5	3445.0	7.5343
780	1.5	4.2133	0.23734	3132.7	3488.7	7.5911
800	1.5	4.1036	0.24369	3167.0	3532.6	7.6466
820	1.5	3.9997	0.25002	3201.6	3576.6	7.701
840	1.5	3.9011	0.25634	3236.4	3620.9	7.7543
860	1.5	3.8075	0.26264	3271.4	3665.3	7.8066
880	1.5	3.7184	0.26893	3306.6	3710.0	7.858
900	1.5	3.6335	0.27522	3342.1	3754.9	7.9084
920	1.5	3.5525	0.28149	3377.8	3800.0	7.958
940	1.5	3.4752	0.28775	3413.8	3845.4	8.0068
960	1.5	3.4013	0.29401	3450.0	3891.0	8.0548
980	1.5	3.3305	0.30026	3486.5	3936.9	8.1021

Table D.3K Isobaric Data for $P = 2.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
485.53	2	10.042	0.099585	2599.1	2798.3	6.339
500	2	9.5781	0.10441	2632.6	2841.4	6.4265
520	2	9.0447	0.11056	2674.0	2895.1	6.5319
540	2	8.5934	0.11637	2712.6	2945.4	6.6267
560	2	8.2008	0.12194	2749.5	2993.4	6.7141
580	2	7.853	0.12734	2785.3	3040.0	6.7959
600	2	7.5406	0.13262	2820.4	3085.6	6.8732
620	2	7.2573	0.13779	2855.0	3130.6	6.947
640	2	6.9983	0.14289	2889.4	3175.1	7.0176
660	2	6.7599	0.14793	2923.5	3219.4	7.0857
680	2	6.5395	0.15292	2957.6	3263.4	7.1514
700	2	6.3346	0.15786	2991.6	3307.4	7.2151
720	2	6.1436	0.16277	3025.8	3351.3	7.277
740	2	5.9648	0.16765	3059.9	3395.2	7.3372
760	2	5.7969	0.1725	3094.2	3439.3	7.3959
780	2	5.639	0.17734	3128.7	3483.4	7.4532
800	2	5.4901	0.18215	3163.3	3527.6	7.5092
820	2	5.3493	0.18694	3198.1	3572.0	7.564
840	2	5.2159	0.19172	3233.1	3616.5	7.6176
860	2	5.0894	0.19649	3268.3	3661.2	7.6702
880	2	4.9691	0.20124	3303.7	3706.1	7.7219
900	2	4.8547	0.20599	3339.3	3751.3	7.7726
920	2	4.7456	0.21072	3375.2	3796.6	7.8224
940	2	4.6415	0.21545	3411.3	3842.2	7.8714
960	2	4.5421	0.22016	3447.6	3887.9	7.9196
980	2	4.4469	0.22488	3484.2	3934.0	7.967
1000	2	4.3558	0.22958	3521.1	3980.2	8.0137

Table D.3L Isobaric Data for $P = 3.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
507	3	15.001	0.066664	2603.2	2803.2	6.1856
520	3	14.309	0.069888	2637.1	2846.7	6.2704
540	3	13.442	0.074391	2682.9	2906.1	6.3825
560	3	12.729	0.078561	2724.7	2960.4	6.4812
580	3	12.119	0.082512	2764.1	3011.6	6.5712
600	3	11.587	0.086307	2801.9	3060.8	6.6546
620	3	11.113	0.089986	2838.7	3108.6	6.733
640	3	10.687	0.093576	2874.7	3155.5	6.8073
660	3	10.299	0.097096	2910.3	3201.6	6.8783
680	3	9.9443	0.10056	2945.6	3247.3	6.9465
700	3	9.6174	0.10398	2980.7	3292.6	7.0122
720	3	9.3147	0.10736	3015.7	3337.7	7.0758
740	3	9.033	0.1107	3050.6	3382.7	7.1374
760	3	8.77	0.11403	3085.6	3427.7	7.1973
780	3	8.5236	0.11732	3120.6	3472.6	7.2557
800	3	8.292	0.1206	3155.8	3517.6	7.3126
820	3	8.0738	0.12386	3191.0	3562.6	7.3682
840	3	7.8678	0.1271	3226.4	3607.7	7.4226
860	3	7.6729	0.13033	3262.0	3653.0	7.4758
880	3	7.488	0.13355	3297.8	3698.4	7.528
900	3	7.3124	0.13675	3333.7	3744.0	7.5792
920	3	7.1454	0.13995	3369.9	3789.7	7.6295
940	3	6.9863	0.14314	3406.2	3835.7	7.6789
960	3	6.8344	0.14632	3442.8	3881.8	7.7275
980	3	6.6893	0.14949	3479.7	3928.1	7.7752
1000	3	6.5506	0.15266	3516.7	3974.7	7.8223
1020	3	6.4177	0.15582	3554.0	4021.5	7.8686

Table D.3M Isobaric Data for $P = 4.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
523.5	4	20.09	0.049776	2601.7	2800.8	6.0696
540	4	18.831	0.053103	2648.4	2860.8	6.1824
560	4	17.642	0.056683	2696.9	2923.7	6.2968
580	4	16.675	0.05997	2740.9	2980.8	6.397
600	4	15.857	0.063063	2782.1	3034.3	6.4878
620	4	15.147	0.066018	2821.4	3085.4	6.5716
640	4	14.52	0.069	2859.4	3134.9	6.6501
660	4	13.958	0.072	2896.6	3183.2	6.7244
680	4	13.449	0.074	2933.2	3230.6	6.7953
700	4	12.985	0.077	2969.4	3277.5	6.8632
720	4	12.558	0.080	3005.4	3323.9	6.9285
740	4	12.163	0.082	3041.1	3370.0	6.9917
760	4	11.796	0.085	3076.8	3415.9	7.0529
780	4	11.454	0.087	3112.5	3461.7	7.1124
800	4	11.134	0.090	3148.2	3507.4	7.1702
820	4	10.833	0.092	3183.9	3553.1	7.2267
840	4	10.55	0.095	3219.7	3598.9	7.2818
860	4	10.283	0.097	3255.7	3644.7	7.3357
880	4	10.03	0.100	3291.8	3690.6	7.3885
900	4	9.791	0.102	3328.1	3736.6	7.4402
920	4	9.5636	0.105	3364.5	3782.8	7.4909
940	4	9.3473	0.10698	3401.2	3829.1	7.5407
960	4	9.1412	0.10939	3438.0	3875.6	7.5897
980	4	8.9446	0.1118	3475.1	3922.3	7.6378
1000	4	8.7568	0.1142	3512.4	3969.1	7.6851
1020	4	8.5771	0.11659	3549.9	4016.2	7.7317
1040	4	8.405	0.11898	3587.6	4063.5	7.7776

Table D.3N Isobaric Data for $P = 6.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
548.73	6	30.818	0.032448	2589.9	2784.6	5.8901
560	6	29.166	0.034287	2629.1	2834.8	5.9807
580	6	26.947	0.03711	2687.2	2909.8	6.1125
600	6	25.247	0.039608	2737.6	2975.2	6.2233
620	6	23.865	0.041903	2783.5	3034.9	6.3211
640	6	22.697	0.044058	2826.5	3090.8	6.4099
660	6	21.686	0.046112	2867.5	3144.2	6.4921
680	6	20.795	0.048089	2907.2	3195.7	6.569
700	6	19.998	0.050006	2945.9	3246.0	6.6418
720	6	19.277	0.051875	2984.0	3295.2	6.7112
740	6	18.62	0.053706	3021.5	3343.8	6.7777
760	6	18.017	0.055504	3058.7	3391.8	6.8417
780	6	17.46	0.057274	3095.7	3439.4	6.9036
800	6	16.943	0.059022	3132.6	3486.7	6.9635
820	6	16.461	0.06075	3169.4	3533.9	7.0217
840	6	16.01	0.062461	3206.1	3580.9	7.0784
860	6	15.587	0.064157	3242.9	3627.9	7.1336
880	6	15.188	0.06584	3279.8	3674.8	7.1876
900	6	14.812	0.067512	3316.7	3721.8	7.2404
920	6	14.457	0.069173	3353.8	3768.8	7.2921
940	6	14.119	0.070825	3391.0	3815.9	7.3427
960	6	13.799	0.072469	3428.4	3863.2	7.3924
980	6	13.494	0.074105	3465.9	3910.5	7.4413
1000	6	13.204	0.075735	3503.6	3958.0	7.4892
1020	6	12.927	0.077358	3541.5	4005.6	7.5364
1040	6	12.662	0.078977	3579.6	4053.4	7.5828
1060	6	12.409	0.08059	3617.9	4101.4	7.6285

Table D.30 Isobaric Data for $P = 8.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
568.16	8	42.507	0.023526	2570.5	2758.7	5.745
580	8	39.647	0.025223	2619.0	2820.7	5.8531
600	8	36.218	0.027611	2684.7	2905.6	5.9971
620	8	33.704	0.02967	2740.2	2977.6	6.1151
640	8	31.713	0.031532	2789.9	3042.1	6.2176
660	8	30.065	0.033261	2835.8	3101.9	6.3096
680	8	28.658	0.034894	2879.3	3158.4	6.394
700	8	27.431	0.036455	2921.0	3212.6	6.4726
720	8	26.343	0.037961	2961.5	3265.2	6.5466
740	8	25.367	0.039422	3001.1	3316.5	6.6168
760	8	24.482	0.040847	3040.0	3366.8	6.6839
780	8	23.673	0.042242	3078.5	3416.4	6.7484
800	8	22.929	0.043612	3116.6	3465.5	6.8105
820	8	22.242	0.044961	3154.5	3514.2	6.8706
840	8	21.602	0.046292	3192.3	3562.6	6.9289
860	8	21.006	0.047606	3229.9	3610.8	6.9856
880	8	20.447	0.048907	3267.6	3658.8	7.0409
900	8	19.922	0.050196	3305.2	3706.8	7.0948
920	8	19.427	0.051475	3342.9	3754.7	7.1474
940	8	18.96	0.052744	3380.7	3802.6	7.199
960	8	18.517	0.054004	3418.6	3850.6	7.2495
980	8	18.097	0.055257	3456.6	3898.6	7.299
1000	8	17.698	0.056503	3494.7	3946.8	7.3476
1020	8	17.318	0.057742	3533.1	3995.0	7.3953
1040	8	16.956	0.058976	3571.5	4043.3	7.4423
1060	8	16.61	0.060205	3610.2	4091.8	7.4885
1080	8	16.279	0.06143	3649.0	4140.5	7.5339

Table D.3P Isobaric Data for $P = 10.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
584.15	10	55.463	0.01803	2545.2	2725.5	5.616
600	10	49.773	0.020091	2619.1	2820.0	5.7756
620	10	45.151	0.022148	2689.7	2911.2	5.9253
640	10	41.84	0.0239	2748.7	2987.7	6.0468
660	10	39.259	0.025472	2801.1	3055.8	6.1516
680	10	37.145	0.026922	2849.2	3118.4	6.2451
700	10	35.355	0.028285	2894.5	3177.4	6.3305
720	10	33.804	0.029582	2937.8	3233.7	6.4098
740	10	32.437	0.030829	2979.7	3288.0	6.4843
760	10	31.215	0.032036	3020.6	3341.0	6.5549
780	10	30.112	0.033209	3060.7	3392.8	6.6222
800	10	29.107	0.034356	3100.2	3443.7	6.6867
820	10	28.185	0.035479	3139.3	3494.1	6.7489
840	10	27.335	0.036584	3178.1	3543.9	6.8089
860	10	26.546	0.037671	3216.7	3593.4	6.8671
880	10	25.81	0.038744	3255.1	3642.6	6.9237
900	10	25.123	0.039804	3293.5	3691.6	6.9787
920	10	24.478	0.040854	3331.9	3740.4	7.0324
940	10	23.87	0.041893	3370.3	3789.2	7.0849
960	10	23.297	0.042924	3408.7	3837.9	7.1362
980	10	22.755	0.043947	3447.2	3886.7	7.1864
1000	10	22.241	0.044963	3485.8	3935.5	7.2357
1020	10	21.752	0.045972	3524.6	3984.3	7.284
1040	10	21.287	0.046976	3563.4	4033.2	7.3315
1060	10	20.844	0.047975	3602.5	4082.2	7.3782
1080	10	20.421	0.048969	3641.6	4131.3	7.4241
1100	10	20.017	0.049959	3681.0	4180.6	7.4693

Table D.3Q Isobaric Data for $P = 12.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
597.83	12	70.106	0.014264	2514.3	2685.4	5.4939
600	12	68.549	0.014588	2528.8	2703.8	5.5247
620	12	59.113	0.016917	2628.8	2831.8	5.7346
640	12	53.502	0.018691	2701.7	2926.0	5.8843
660	12	49.499	0.020202	2762.6	3005.0	6.0059
680	12	46.392	0.021555	2816.6	3075.3	6.1108
700	12	43.857	0.022801	2866.3	3139.9	6.2045
720	12	41.718	0.023971	2912.9	3200.5	6.2899
740	12	39.87	0.025082	2957.4	3258.4	6.3692
760	12	38.245	0.026147	3000.4	3314.2	6.4436
780	12	36.796	0.027177	3042.3	3368.4	6.514
800	12	35.49	0.028177	3083.3	3421.4	6.5811
820	12	34.303	0.029152	3123.7	3473.5	6.6454
840	12	33.215	0.030107	3163.6	3524.9	6.7073
860	12	32.213	0.031044	3203.2	3575.7	6.7671
880	12	31.284	0.031966	3242.5	3626.1	6.8251
900	12	30.419	0.032874	3281.7	3676.2	6.8813
920	12	29.611	0.033771	3320.7	3726.0	6.9361
940	12	28.853	0.034658	3359.7	3775.6	6.9895
960	12	28.14	0.035536	3398.7	3825.2	7.0416
980	12	27.468	0.036406	3437.8	3874.6	7.0926
1000	12	26.832	0.037269	3476.9	3924.1	7.1425
1020	12	26.229	0.038126	3516.0	3973.5	7.1915
1040	12	25.657	0.038976	3555.3	4023.0	7.2395
1060	12	25.112	0.039821	3594.7	4072.5	7.2867
1080	12	24.593	0.040662	3634.2	4122.1	7.3331
1100	12	24.098	0.041498	3673.9	4171.8	7.3787

Table D.3R Isobaric Data for $P = 14.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
609.82	14	87.069	0.011485	2477.1	2637.9	5.3727
620	14	77.66	0.012877	2549.9	2730.1	5.5229
640	14	67.43	0.01483	2646.6	2854.2	5.72
660	14	61.128	0.016359	2719.6	2948.6	5.8653
680	14	56.586	0.017672	2781.1	3028.5	5.9847
700	14	53.047	0.018851	2836.0	3099.9	6.0882
720	14	50.153	0.019939	2886.5	3165.7	6.1808
740	14	47.711	0.020959	2934.1	3227.5	6.2655
760	14	45.602	0.021929	2979.5	3286.5	6.3441
780	14	43.747	0.022859	3023.3	3343.3	6.418
800	14	42.094	0.023756	3066.0	3398.5	6.4879
820	14	40.605	0.024628	3107.7	3452.5	6.5545
840	14	39.252	0.025477	3148.8	3505.5	6.6184
860	14	38.012	0.026307	3189.5	3557.8	6.6799
880	14	36.871	0.027122	3229.7	3609.4	6.7392
900	14	35.813	0.027923	3269.7	3660.6	6.7967
920	14	34.829	0.028712	3309.5	3711.4	6.8526
940	14	33.91	0.02949	3349.1	3762.0	6.9069
960	14	33.048	0.030259	3388.7	3812.3	6.96
980	14	32.237	0.03102	3428.2	3862.5	7.0117
1000	14	31.472	0.031774	3467.8	3912.6	7.0623
1020	14	30.749	0.032521	3507.4	3962.7	7.1119
1040	14	30.064	0.033262	3547.1	4012.8	7.1605
1060	14	29.414	0.033998	3586.8	4062.8	7.2082
1080	14	28.795	0.034729	3626.7	4112.9	7.255
1100	14	28.205	0.035455	3666.7	4163.1	7.301
1120	14	27.641	0.036178	3706.8	4213.3	7.3463

Table D.3S Isobaric Data for $P = 16.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
620.5	16	107.42	0.009309	2431.8	2580.8	5.2463
640	16	85.058	0.011757	2579.1	2767.2	5.5427
660	16	74.683	0.01339	2670.5	2884.8	5.7237
680	16	67.984	0.014709	2742.1	2977.5	5.8622
700	16	63.065	0.015857	2803.5	3057.2	5.9778
720	16	59.195	0.016893	2858.6	3128.9	6.0788
740	16	56.014	0.017853	2909.6	3195.3	6.1697
760	16	53.32	0.018755	2957.7	3257.8	6.253
780	16	50.988	0.019613	3003.7	3317.5	6.3306
800	16	48.935	0.020435	3048.1	3375.1	6.4035
820	16	47.103	0.021230	3091.4	3431.1	6.4726
840	16	45.452	0.022001	3133.8	3485.8	6.5386
860	16	43.951	0.022753	3175.5	3539.5	6.6018
880	16	42.576	0.023488	3216.7	3592.5	6.6627
900	16	41.308	0.024208	3257.5	3644.8	6.7215
920	16	40.134	0.024916	3298.0	3696.7	6.7785
940	16	39.042	0.025614	3338.3	3748.2	6.8338
960	16	38.021	0.026301	3378.5	3799.4	6.8877
980	16	37.063	0.026981	3418.6	3850.3	6.9403
1000	16	36.163	0.027653	3458.7	3901.1	6.9916
1020	16	35.313	0.028318	3498.8	3951.8	7.0418
1040	16	34.51	0.028977	3538.8	4002.5	7.091
1060	16	33.749	0.029631	3579.0	4053.1	7.1391
1080	16	33.026	0.030279	3619.2	4103.7	7.1864
1100	16	32.338	0.030924	3659.5	4154.3	7.2329
1120	16	31.682	0.031564	3700.0	4205.0	7.2785
1140	16	31.056	0.0322	3740.5	4255.7	7.3235

Table D.3T Isobaric Data for $P = 18.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
630.14	18	133.3	0.0075017	2374.8	2509.8	5.1061
640	18	110	0.0090911	2489.2	2652.8	5.3314
660	18	91.08	0.010979	2613.3	2810.9	5.5749
680	18	80.955	0.012353	2698.9	2921.2	5.7397
700	18	74.095	0.013496	2768.4	3011.3	5.8704
720	18	68.943	0.014505	2829.0	3090.1	5.9813
740	18	64.839	0.015423	2883.9	3161.6	6.0793
760	18	61.439	0.016276	2935.0	3228.0	6.1679
780	18	58.544	0.017081	2983.4	3290.9	6.2495
800	18	56.029	0.017848	3029.8	3351.0	6.3257
820	18	53.809	0.018584	3074.7	3409.2	6.3975
840	18	51.825	0.019296	3118.4	3465.7	6.4656
860	18	50.034	0.019986	3161.3	3521.0	6.5307
880	18	48.403	0.02066	3203.5	3575.3	6.5931
900	18	46.908	0.021318	3245.2	3628.9	6.6533
920	18	45.529	0.021964	3286.5	3681.8	6.7115
940	18	44.251	0.022598	3327.5	3734.3	6.7679
960	18	43.06	0.023223	3368.3	3786.3	6.8227
980	18	41.948	0.023839	3409.0	3838.1	6.876
1000	18	40.904	0.024448	3449.5	3889.6	6.9281
1020	18	39.922	0.025049	3490.0	3940.9	6.9789
1040	18	38.995	0.025645	3530.6	3992.2	7.0286
1060	18	38.118	0.026234	3571.1	4043.3	7.0774
1080	18	37.287	0.026819	3611.7	4094.4	7.1251
1100	18	36.497	0.0274	3652.3	4145.5	7.172
1120	18	35.745	0.027976	3693.1	4196.6	7.2181
1140	18	35.029	0.028548	3733.9	4247.8	7.2633

Table D.3U Isobaric Data for $P = 20.0$ MPa

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
638.9	20	170.5	0.005865	2295.0	2412.3	4.9314
640	20	160.5	0.006231	2328.4	2453.0	4.995
660	20	112.09	0.008921	2543.7	2722.1	5.4104
680	20	96.059	0.01041	2650.1	2858.3	5.6139
700	20	86.38	0.011577	2730.2	2961.8	5.7639
720	20	79.525	0.012575	2797.4	3048.9	5.8867
740	20	74.257	0.013467	2857.0	3126.3	5.9927
760	20	70.002	0.014285	2911.5	3197.2	6.0872
780	20	66.444	0.01505	2962.5	3263.5	6.1733
800	20	63.396	0.015774	3010.9	3326.4	6.253
820	20	60.736	0.016465	3057.5	3386.8	6.3276
840	20	58.379	0.017129	3102.7	3445.3	6.398
860	20	56.268	0.017772	3146.8	3502.2	6.465
880	20	54.357	0.018397	3190.1	3558.0	6.5291
900	20	52.615	0.019006	3232.7	3612.8	6.5907
920	20	51.015	0.019602	3274.8	3666.8	6.6501
940	20	49.538	0.020186	3316.5	3720.2	6.7075
960	20	48.168	0.020761	3358.0	3773.2	6.7633
980	20	46.891	0.021326	3399.2	3825.7	6.8174
1000	20	45.696	0.021884	3440.3	3878.0	6.8702
1020	20	44.574	0.022435	3481.3	3930.0	6.9217
1040	20	43.518	0.022979	3522.2	3981.8	6.972
1060	20	42.521	0.023518	3563.2	4033.5	7.0213
1080	20	41.577	0.024052	3604.1	4085.1	7.0695
1100	20	40.682	0.024581	3645.1	4136.7	7.1168
1120	20	39.831	0.025106	3686.1	4188.3	7.1633
1140	20	39.021	0.025627	3727.3	4239.8	7.2089

Table D.3V Isobaric Data for $P = 26.0$ MPa (Supercritical)

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
660	26	365.88	0.002733	2010.9	2082.0	4.386
680	26	167.08	0.005985	2448.8	2604.4	5.1692
700	26	134.79	0.007419	2591.2	2784.1	5.43
720	26	118.02	0.008473	2688.8	2909.1	5.6062
740	26	106.98	0.009348	2767.2	3010.2	5.7447
760	26	98.863	0.010115	2834.6	3097.6	5.8613
780	26	92.512	0.010809	2895.3	3176.3	5.9635
800	26	87.325	0.011451	2951.2	3248.9	6.0554
820	26	82.963	0.012054	3003.7	3317.1	6.1396
840	26	79.211	0.012625	3053.7	3382.0	6.2178
860	26	75.927	0.01317	3101.9	3444.3	6.2912
880	26	73.015	0.013696	3148.6	3504.7	6.3606
900	26	70.403	0.014204	3194.2	3563.5	6.4267
920	26	68.039	0.014697	3239.0	3621.1	6.4899
940	26	65.882	0.015179	3283.0	3677.6	6.5507
960	26	63.902	0.015649	3326.5	3733.3	6.6094
980	26	62.074	0.01611	3369.5	3788.4	6.6661
1000	26	60.378	0.016562	3412.2	3842.9	6.7211
1020	26	58.797	0.017008	3454.7	3896.9	6.7747
1040	26	57.318	0.017447	3497.0	3950.6	6.8268
1060	26	55.93	0.01788	3539.2	4004.0	6.8777
1080	26	54.623	0.018307	3581.2	4057.2	6.9274
1100	26	53.389	0.01873	3623.3	4110.2	6.976
1120	26	52.222	0.019149	3665.3	4163.1	7.0237
1140	26	51.115	0.019564	3707.3	4216.0	7.0704
1160	26	50.062	0.019975	3749.4	4268.7	7.1163
1180	26	49.06	0.020383	3791.5	4321.4	7.1614
1200	26	48.104	0.020788	3833.7	4374.2	7.2057

Table D.3W Isobaric Data for $P = 28.0$ MPa (Supercritical)

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
660	28	455.47	0.0021955	1897.4	1958.9	4.1923
680	28	210.73	0.0047453	2347.1	2480.0	4.9705
700	28	157.04	0.0063676	2533.7	2712.0	5.3072
720	28	133.92	0.0074672	2647.0	2856.1	5.5104
740	28	119.74	0.0083511	2733.9	2967.7	5.6634
760	28	109.74	0.0091127	2806.9	3062.0	5.7892
780	28	102.1	0.0097942	2871.3	3145.6	5.8977
800	28	95.978	0.010419	2930.1	3221.9	5.9943
820	28	90.896	0.011002	2984.9	3293.0	6.0821
840	28	86.571	0.011551	3036.8	3360.2	6.1632
860	28	82.818	0.012075	3086.5	3424.6	6.2389
880	28	79.511	0.012577	3134.5	3486.6	6.3102
900	28	76.563	0.013061	3181.1	3546.8	6.3779
920	28	73.906	0.013531	3226.8	3605.6	6.4425
940	28	71.494	0.013987	3271.6	3663.2	6.5044
960	28	69.286	0.014433	3315.8	3719.9	6.5641
980	28	67.254	0.014869	3359.5	3775.8	6.6217
1000	28	65.374	0.015297	3402.8	3831.1	6.6775
1020	28	63.626	0.015717	3445.8	3885.8	6.7318
1040	28	61.994	0.016131	3488.5	3940.2	6.7845
1060	28	60.465	0.016538	3531.1	3994.2	6.8359
1080	28	59.029	0.016941	3573.6	4047.9	6.8862
1100	28	57.675	0.017339	3615.9	4101.4	6.9353
1120	28	56.395	0.017732	3658.3	4154.8	6.9833
1140	28	55.183	0.018121	3700.6	4208.0	7.0304
1160	28	54.033	0.018507	3743.0	4261.2	7.0767
1180	28	52.938	0.01889	3785.3	4314.3	7.122
1200	28	51.895	0.01927	3827.8	4367.3	7.1666

Table D.3X Isobaric Data for $P = 32.0$ MPa (Supercritical)

T (K)	P (MPa)	ρ (kg/m ³)	v (m ³ /kg)	u (kJ/kg)	h (kJ/kg)	s (kJ/kg·K)
660	32	516.64	0.001936	1823.6	1885.6	4.0689
680	32	351.66	0.002844	2100.2	2191.2	4.5242
700	32	217.87	0.00459	2395.5	2542.3	5.0338
720	32	172.38	0.005801	2553.3	2739.0	5.3111
740	32	148.9	0.006716	2661.8	2876.7	5.4999
760	32	133.76	0.007476	2747.9	2987.1	5.6471
780	32	122.84	0.00814	2821.2	3081.7	5.77
800	32	114.42	0.00874	2886.5	3166.2	5.877
820	32	107.62	0.009292	2946.2	3243.6	5.9726
840	32	101.96	0.009808	3002.0	3315.9	6.0597
860	32	97.13	0.010295	3054.9	3384.4	6.1403
880	32	92.934	0.01076	3105.6	3449.9	6.2156
900	32	89.235	0.011206	3154.5	3513.1	6.2866
920	32	85.935	0.011637	3202.1	3574.4	6.354
940	32	82.962	0.012054	3248.6	3634.3	6.4184
960	32	80.262	0.012459	3294.3	3692.9	6.4802
980	32	77.791	0.012855	3339.3	3750.6	6.5396
1000	32	75.517	0.013242	3383.7	3807.5	6.597
1020	32	73.413	0.013622	3427.7	3863.6	6.6527
1040	32	71.458	0.013994	3471.5	3919.3	6.7067
1060	32	69.633	0.014361	3514.9	3974.4	6.7592
1080	32	67.924	0.014722	3558.1	4029.3	6.8105
1100	32	66.318	0.015079	3601.2	4083.8	6.8605
1120	32	64.804	0.015431	3644.3	4138.0	6.9094
1140	32	63.374	0.015779	3687.2	4192.1	6.9572
1160	32	62.019	0.016124	3730.1	4246.1	7.0041
1180	32	60.734	0.016465	3773.0	4299.9	7.0502
1200	32	59.511	0.016804	3816.0	4353.7	7.0953